

Statistical Questions and How Spread Out the Data Is Answer Key

Grade 6–7

Quick Answers

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|---------------|---------------|------------------|
| 1. 0 | 5. 45 | 9. $\frac{2}{5}$ |
| 2. 22 | 6. 30 | 10. — |
| 3. 0, 2, 0, 5 | 7. 20, 40, 20 | |
| 4. 32 | 8. 12, 5, 33 | |
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Curriculum

Level: Grade 6–7

6.SP.A – *problems 1, 2, 3, 10*

6.SP.B.5 – *problems 4, 5, 6, 7, 8*

7.SP.C.5, 7.SP.C.7 – *problem 9*

Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

6. *If they got 35: took the middle value of the unsorted list instead of sorting first*

What this sheet is teaching. Two ideas, and the second is the one students skip. First, a question is statistical when its answers vary — which is a property of the *answers*, not of the topic, so “How tall is Priya?” is not statistical even though it is about height. Second, a distribution needs two numbers to describe it, a centre and a spread, and the sheet ends by putting two classes with almost identical means side by side so the student can see what the mean alone leaves out.

1. Not every question is statistical. Answers to “How tall is Priya?”: 148, 148, 148, 148.

Largest 148, smallest 148, so the range is $148 - 148 = \boxed{0}$.

Every dot sits in one stack. The answers do not vary at all, so this is **not** a statistical question — Priya has one height, and asking four people cannot change that.

The point. Being about a measurable quantity is not enough. Variability in the answers is the whole test, and a range of zero is what “no variability” looks like as a number.

2. Now change one word. Answers to “How tall are the students in our class?”: 140, 146, 151, 155, 162.

Largest 162, smallest 140, so the range is $162 - 140 = \boxed{22}$ cm.

The dots spread right across the number line. The answers vary, so this **is** a statistical question.

The word that changed. “Priya” became “the students in our class” — a single individual became a group. A statistical question asks about a group, precisely because a group is what produces a spread of answers.

3. Sort the four questions.

question	range	statistical?
(a) How old is our teacher?	$\boxed{0}$	no
(b) How old are the students in Grade 6?	$\boxed{2}$	yes
(c) How many pets does Sam have?	$\boxed{0}$	no
(d) How many pets do the students in our class have?	$\boxed{5}$	yes

The rule in the student’s own words should be something like: “if the answers can be different from each other, it is statistical”. Both of the non-statistical questions name *one* person; both of the statistical ones name a *group*.

Worth noticing about (b). Its range is only 2 years — small, but not zero. A question does not have to produce a big spread to be statistical; it only has to produce some.

4. Class A: find the centre. Data: 15, 20, 20, 25, 30, 30, 35, 40, 45, 60.

Sum: $15 + 20 + 20 + 25 + 30 + 30 + 35 + 40 + 45 + 60 = \boxed{320}$.

Ten values, so the mean is $320 \div 10 = \boxed{32}$ minutes.

A useful sanity check. Only three of the ten students spent more than 32 minutes, yet 32 is the “average”. That is the single value of 60 pulling the mean upward, and it is the reason problem 6 asks for the median as well.

5. Class A: find the range.

Largest $\boxed{60}$, smallest $\boxed{15}$, so the range is $60 - 15 = \boxed{45}$ minutes.

What the range adds. The mean says a typical night was about half an hour. The range says the class’s experience of that same homework ran from a quarter of an hour to a full hour — a four-fold difference between the fastest and the slowest student. A teacher setting homework needs the second fact at least as much as the first.

6. Class A: find the median — carefully.

What Sam did. Sam took the middle of the list *as written* rather than of the sorted values, and landed on 35. The median is defined on sorted data; without sorting, the “middle of the list” is just whichever value happened to be typed fifth.

Sorted: 15, 20, 20, 25, 30, 30, 35, 40, 45, 60. With ten values the median is halfway between the 5th and the 6th:

$$5\text{th} = \boxed{30}, \quad 6\text{th} = 30, \quad \text{median} = \frac{30 + 30}{2} = \boxed{30} \text{ minutes.}$$

Mean 32, median 30. The gap between them is small here but it points the right way: the mean sits above the median because the distribution has a tail to the right. Ask the student which of the two they would quote to a parent who wanted to know what a “normal” night looks like.

7. Class A: the middle half.

Sorted, split into halves of five:

$$\underbrace{15, 20, \boxed{20}, 25, 30}_{\text{lower half}} \quad | \quad \underbrace{30, 35, \boxed{40}, 45, 60}_{\text{upper half}}$$

$Q_1 = \boxed{20}$ (median of the lower half), $Q_3 = \boxed{40}$ (median of the upper half).

$\text{IQR} = Q_3 - Q_1 = 40 - 20 = \boxed{20}$ minutes.

Why bother, when the range was easier? The range of 45 is decided entirely by the one student who spent 60 minutes. Delete that student and the range collapses to 30; the IQR does not move at all, because the middle five values are untouched. The IQR describes the bulk of the class, the range describes its extremes, and both are honest answers to different questions.

8. Class B, and a comparison. Data: 28, 30, 30, 32, 32, 33, 34, 35, 36, 40.

Range. Largest 40, smallest 28: range = $40 - 28 = \boxed{12}$ minutes.

Quartiles. Lower half 28, 30, 30, 32, 32 gives $Q_1 = 30$; upper half 33, 34, 35, 36, 40 gives $Q_3 = 35$. So $\text{IQR} = 35 - 30 = \boxed{5}$ minutes.

The comparison. Class A’s range is 45 and Class B’s is 12, a difference of $45 - 12 = \boxed{33}$ minutes.

Which class is easier to predict? Class B, and the numbers say why: half of Class B sits inside a five-minute window, while half of Class A is spread over twenty minutes. If you had to guess how long one more student would take, you would be far more often right about Class B.

9. One student picked at random.

The median is 30 minutes. Strictly more than 30: the values 35, 40, 45, 60 — $\boxed{4}$ students out of $\boxed{10}$.

$$P(\text{more than 30 minutes}) = \frac{4}{10} = \boxed{\frac{2}{5}}$$

Why it is not exactly one half. Two students landed exactly *on* 30, and “more than 30” excludes them. With repeated values a median does not have to split the data into two equal groups — a detail worth surfacing, because students often assume the answer to this kind of question must be $\frac{1}{2}$.

10. Write to the teacher. (Open explanation — read the reasoning.)

The facts to be used. Class A: mean 32, range 45, IQR 20. Class B: mean 33, range 12, IQR 5. The centres differ by one minute; the spreads differ by a factor of about four.

A model answer. “Both classes averaged about half an hour, so on the mean alone they look identical. But Class A’s homework times ran from 15 minutes to an hour — a range of 45 — while Class B’s ran from 28 to 40, a range of only 12. The middle half of Class A covers 20 minutes and the middle half of Class B covers 5. So in Class B almost everyone had a similar night, and in Class A some students finished in a quarter of an hour while others spent four times that. I would report the median and the IQR for each class, not just the mean, because the mean hides which students the homework is actually hard for.”

Marking guide. Full credit needs a spread measure quoted with its number and used to separate the two classes. Partial credit for saying the classes differ “in how spread out they are” without quoting a range or IQR. Not sufficient: repeating that the means are 32 and 33, or saying Class A “has an outlier” without saying what that does to the description.

A better-than-asked-for answer notices that the two classes might need different responses from the teacher: Class B’s homework is well calibrated, while Class A contains students for whom the same task takes four times as long.

Open explanation — open response.